A simple proof of the Calabi-Bernstein theorem for entire maximal graphs

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Abstract

The fundamental Calabi-Bernstein theorem states that the only entire solutions to the maximal surface equation in the three-dimensional Lorentz-Minkowski space $\mathbb{L}^3$ are affine functions. Geometrically, this asserts that the only entire maximal graphs in $\mathbb{L}^3$ are spacelike planes. While classical proofs rely on techniques from complex analysis and holomorphic representations, A. Romero [7] formulated a purely differential-geometric proof. In this paper, we expand and detail Romero's approach. We meticulously establish the required semi-Riemannian geometry concepts, demonstrate the construction of a complete and flat conformal metric on the entire graph, and show how the uniformization theorem coupled with Liouville's theorem for harmonic functions smoothly yields the result.

1 Introduction

The study of hypersurfaces with prescribed mean curvature stands as one of the pillars of differential geometry. Among the most celebrated results in this area is the classical Bernstein theorem, established by S. N. Bernstein in 1915. Bernstein demonstrated that any entire minimal graph in the three-dimensional Euclidean space $\mathbb{R}^3$ must necessarily be a plane. In the language of partial differential equations, Bernstein's theorem asserts that the only entire solutions $u(x,y)$ defined on all of $\mathbb{R}^2$ to the minimal surface equation

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = 0 \tag{1}$$

are affine functions of the form $u(x,y) = ax + by + c$. This rigidity property motivated a vast amount of subsequent research. While classical proofs often rely on complex analysis, S.S. Chern [3] provided a renowned, purely geometric proof of this fact by introducing a flat conformal metric and applying Liouville's theorem to the Laplacian.

Given the success in Euclidean space, mathematicians naturally sought analogous results in the ambient space of special relativity: the Lorentz-Minkowski space $\mathbb{L}^3$. In this Lorentzian setting, the area functional is bounded from above for compactly supported variations, meaning the critical points of the area functional maximize the area, hence the name *maximal surfaces*. The corresponding partial differential equation for a spacelike graph given by $x_3 = u(x_1, x_2)$ is the maximal surface equation:

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1-|\nabla u|^2}}\right) = 0, \quad \text{with } |\nabla u| < 1. \tag{2}$$

The constraint $|\nabla u| < 1$ is geometrically crucial; it ensures that the graph remains strictly *spacelike*, meaning its tangent planes are entirely composed of spacelike vectors[1].

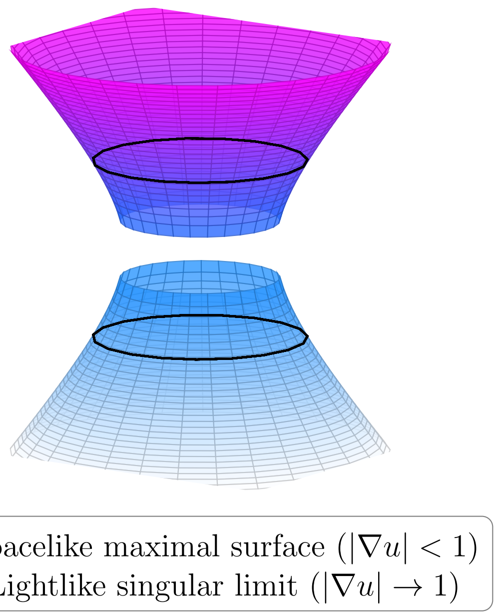

Figure 1: A maximal catenoid in $\mathbb{L}^3$. Notice how the surface cannot be extended as an entire graph over $\mathbb{R}^2$ because its tangent planes eventually become perpendicular to the spatial plane (lightlike), violating the strict spacelike condition. The dashed black rings highlight the threshold where the gradient magnitude $|\nabla u|$ approaches the speed of light.

Eugenio Calabi [1] resolved the Bernstein problem for maximal surfaces in $\mathbb{L}^{n+1}$ for $n \leq 4$, and shortly after, Cheng and Yau [2] proved the theorem for all dimensions n. This result is known as the Calabi-Bernstein theorem.

Original proofs of the Calabi-Bernstein theorem for $\mathbb{L}^3$ often relied heavily on complex representation formulas or on hard analytic estimates for the gradient [6]. However, A. Romero [7] discovered that complex function theory is not strictly necessary for the three-dimensional case. His method avoids producing the Gauss map as a meromorphic function. Instead, he skillfully utilizes the second fundamental form, Laplace operators on the surface, conformal changes of the metric, Cartan's theorem for complete flat manifolds, and the classical Liouville theorem for harmonic functions in the plane.

The purpose of this work is to meticulously rebuild Romero's proof, filling in the standard semi-Riemannian computations and lemmas that are often omitted in research papers. We will progress through the foundations of Lorentz-Minkowski geometry, detail the behavior of the Laplacian on the normal vector field, construct the necessary conformal metric, and conclude the theorem.

[1] The "speed of light" is not exceeded

2 Preliminaries in Lorentz-Minkowski Geometry

To provide a formally complete proof, we must first establish the background of semi-Riemannian geometry, specifically tailored to the three-dimensional Lorentz-Minkowski space and its spacelike surfaces.

2.1 The Ambient Space $\mathbb{L}^3$

Let $\mathbb{R}^3$ be the standard three-dimensional real vector space. We define the Lorentz-Minkowski inner product $\langle \cdot, \cdot \rangle$ of two vectors $v = (v_1, v_2, v_3)$ and $w = (w_1, w_2, w_3)$ by:

$$\langle v, w \rangle = v_1 w_1 + v_2 w_2 - v_3 w_3. \tag{3}$$

The pair $(\mathbb{R}^3, \langle \cdot, \cdot \rangle)$ is denoted as $\mathbb{L}^3$. Unlike the Euclidean inner product, this metric is indefinite. As a consequence, non-zero vectors in $\mathbb{L}^3$ fall into three distinct causal categories:

1. **Spacelike:** If $\langle v, v \rangle > 0$.
2. **Timelike:** If $\langle v, v \rangle < 0$.
3. **Lightlike (or Null):** If $\langle v, v \rangle = 0$ and $v \neq 0$.

The set of all lightlike vectors forms the *light cone*. A timelike vector v is said to be *future-directed* if $v_3 > 0$, and *past-directed* if $v_3 < 0$.

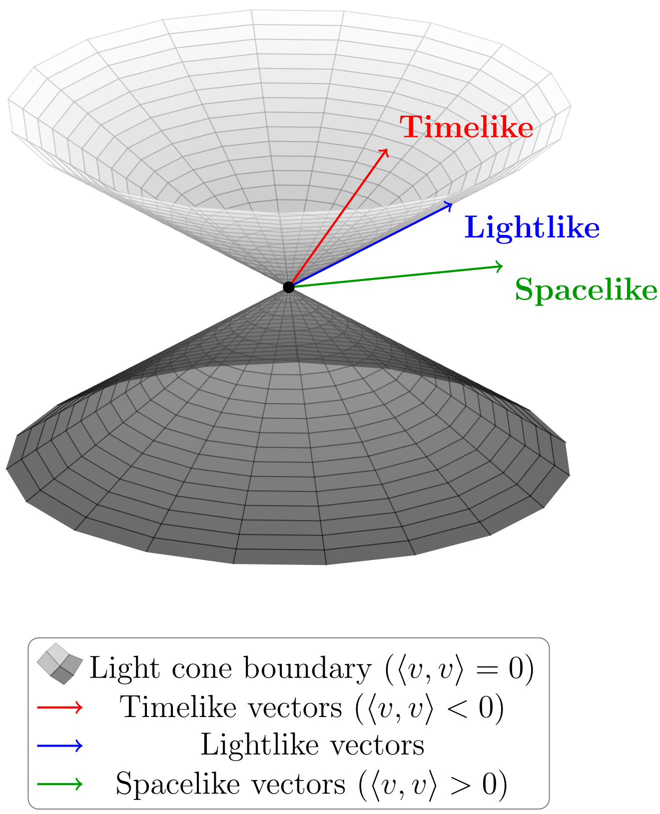

Figure 2: Causal structure of the Lorentz-Minkowski space $\mathbb{L}^3$. The light cone separates the space into timelike (inside), lightlike (boundary), and spacelike (outside) vectors.

A fundamental inequality in $\mathbb{L}^3$ is the *reversed Cauchy-Schwarz inequality.* If v and w are two timelike vectors lying in the same time-cone (e.g., both future-directed), then:

$$\langle v, w\rangle \leq -\sqrt{-\langle v, v\rangle}\sqrt{-\langle w, w\rangle}. \tag{4}$$

2.2 Spacelike Surfaces and the Shape Operator

Let M be a connected, two-dimensional smooth manifold, and let $x : M \to \mathbb{L}^3$ be an immersion. The induced metric on M is the pullback of the Lorentzian metric, defined for tangent vectors $X, Y \in T_pM$ as $g(X, Y) = \langle dx_p(X), dx_p(Y)\rangle$. In local coordinates (x_1, x_2), the induced metric is given by the coefficients $g_{ij} = \langle \partial_i, \partial_j\rangle$, where $\partial_i = \frac{\partial x}{\partial x_i}$.

Definition 1. *The immersion x is called a* ***spacelike surface*** *if the induced metric g is positive definite everywhere on M. Consequently, (M, g) becomes a Riemannian manifold.*

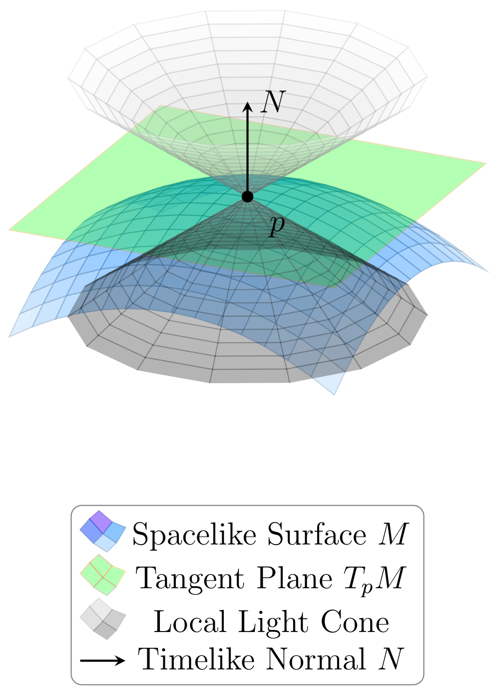

Figure 3: Spacelike surface and its tangent plane at a point p. The light cone at p does not intersect the tangent plane except at the vertex, demonstrating the spacelike nature. The normal vector N is purely timelike and future-directed, lying inside the upper light cone.

Since M is spacelike, its tangent plane T_pM exclusively contains spacelike vectors. The normal line $(T_pM)^\perp$ must therefore be spanned by a timelike vector. Because $\mathbb{L}^3$ is time-orientable, any spacelike surface M admits a unique globally defined, future-directed, unit timelike normal vector field N. That is, $\langle N, N\rangle = -1$ and $N_3 > 0$ everywhere on M. This global non-vanishing vector field also guarantees that M is orientable.

Let $\overline{\nabla}$ denote the flat Levi-Civita connection of $\mathbb{L}^3$, and ∇ the Levi-Civita connection of (M, g). The Gauss and Weingarten formulas relate these connections:

$$\overline{\nabla}_X Y = \nabla_X Y + h(X, Y), \tag{5}$$

$$\overline{\nabla}_X N = -A(X), \tag{6}$$

where X, Y are vector fields tangent to M, h is the second fundamental form, and the map $A : T_pM \to T_pM$ is the Weingarten endomorphism (or shape operator). Note the choice of signs: since $\langle N, N\rangle = -1$, differentiating gives $\langle \overline{\nabla}_X N, N\rangle = 0$, meaning $\overline{\nabla}_X N$ is tangent to M, allowing us to define $A(X) = -\overline{\nabla}_X N$.

In local coordinates, we can decompose the second derivative of the immersion. We define the coefficients of the second fundamental form as $h_{ij} = \langle \partial_{ij}, N\rangle$, noting that $\partial_{ij} = \overline{\nabla}_{\partial_i}\partial_j$. The Weingarten map acts as $A(\partial_i) = \sum_j A_i^j \partial_j$, and the coefficients relate via $h_{ij} = \langle A(\partial_i), \partial_j\rangle = \sum_k A_i^k g_{kj}$. Thus, $A_i^j = \sum_k h_{ik} g^{kj}$, where g^{kj} is the inverse matrix of g_{kj}.

The second fundamental form and the shape operator are related by:

$$\langle A(X), Y\rangle = -\langle \overline{\nabla}_X N, Y\rangle = \langle N, \overline{\nabla}_X Y\rangle = \langle N, h(X, Y)\rangle. \tag{7}$$

Definition 2. *The* ***mean curvature*** *H of M is defined as $H = -\frac{1}{2}tr(A) = -\frac{1}{2}\sum_{i,j} g^{ij}h_{ij}$. A spacelike surface M is called* ***maximal*** *if its mean curvature vanishes identically, i.e., $H \equiv 0$ on M, or equivalently, $tr(A) = 0$.*

The term "maximal" originates from the fact that variations of the surface along the normal direction yield $A'(0) = -2\int_M HfdA$, where f is a compactly supported variation function. If $H = 0$, the surface is a critical point for the area functional, and due to the Lorentzian signature, this critical point actually maximizes the area.

Another important invariant is the Gaussian curvature K of M. By the Gauss equation for submanifolds in a flat ambient space, we have for an orthonormal frame $\{e_1, e_2\}$:

$$K = \langle h(e_1, e_1), N\rangle\langle h(e_2, e_2), N\rangle - \langle h(e_1, e_2), N\rangle^2\langle N, N\rangle. \tag{8}$$

Since $\langle N, N\rangle = -1$, the geometric signature alters the standard Euclidean Gauss equation, leading to $K = -\det(A)$. For a maximal surface, $\mathrm{tr}(A) = 0$, meaning its eigenvalues are λ and $-\lambda$. Thus, $\mathrm{tr}(A^2) = \lambda^2 + (-\lambda)^2 = 2\lambda^2$, and $\det(A) = -\lambda^2$. Therefore, we obtain the pivotal relation:

$$K = \frac{1}{2}\mathrm{tr}(A^2) \geq 0. \tag{9}$$

3 Essential Analytic Identities

Romero's proof relies profoundly on the analytical behavior of functions defined as the inner product of the normal vector N with fixed constant vectors in $\mathbb{L}^3$. We dedicate this section to proving the necessary identities.

Let $a \in \mathbb{L}^3$ be an arbitrary constant vector field. The vector a can be decomposed at any point $p \in M$ into its tangential component $a^\top \in T_pM$ and its normal component $a^\perp \in (T_pM)^\perp$:

$$a = a^\top - \langle N, a\rangle N. \tag{10}$$

Let us define the smooth function $f : M \to \mathbb{R}$ by $f(p) = \langle N_p, a\rangle$. For this function, we will calculate its gradient ∇f and Laplacian Δf.

Lemma 1. *The gradient of the function $f(p) = \langle N_p, a\rangle$ is given by $\nabla\langle N, a\rangle = -A(a^\top)$.*

Proof. By the definition of the gradient on a Riemannian manifold (M, g), for any tangent vector field $X \in \mathfrak{X}(M)$, we have $g(\nabla f, X) = X(f)$. Computing the directional derivative:

$$X\langle N, a\rangle = \langle \overline{\nabla}_X N, a\rangle + \langle N, \overline{\nabla}_X a\rangle.$$

Since a is a constant vector field in $\mathbb{L}^3$, $\overline{\nabla}_X a = 0$. By the Weingarten formula (6), $\overline{\nabla}_X N = -A(X)$. Therefore:

$$X\langle N, a\rangle = \langle -A(X), a\rangle = \langle -A(X), a^\top\rangle.$$

Since the shape operator A is self-adjoint, we obtain:

$$g(\nabla f, X) = g(-X, A(a^\top)) = g(-A(a^\top), X).$$

Since this holds for all X, we conclude $\nabla\langle N, a\rangle = -A(a^\top)$. $\square$

Lemma 2. *If M is a maximal surface ($H = 0$), then the Laplacian of $f = \langle N, a\rangle$ satisfies:*

$$\Delta\langle N, a\rangle = tr(A^2)\langle N, a\rangle. \tag{11}$$

Proof. Following the calculations presented in class, we first compute the Hessian of $f(p) = \langle N, a\rangle$. For tangent vector fields $X, Y \in \mathfrak{X}(M)$, the definition of the Hessian gives $\mathrm{Hess}(f)(X, Y) = X(Y(f)) - (\nabla_X Y)f$. From Lemma 1, we know that

$$Y(f) = g(\nabla f, Y) = \langle -A(Y), a\rangle.$$

Differentiating this with respect to X, we obtain:

$$X\langle -A(Y), a\rangle = -\langle \overline{\nabla}_X(A(Y)), a\rangle = -\langle (\nabla_X A)Y + A(\nabla_X Y) + h(X, A(Y)), a\rangle.$$

The term $(\nabla_X Y)f$ expands to $\langle -A(\nabla_X Y), a\rangle$. Subtracting this from $X(Y(f))$ yields:

$$\mathrm{Hess}(f)(X, Y) = -\langle (\nabla_X A)Y, a\rangle - \langle h(X, A(Y)), a\rangle.$$

By the property of the second fundamental form,

$$h(X, A(Y)) = g(A(X), A(Y))N = -\langle A^2(X), Y\rangle N.$$

Thus, taking the inner product with a, the second term becomes $\langle A^2 X, Y\rangle\langle N, a\rangle$. Substituting this back gives the Hessian:

$$\mathrm{Hess}(f)(X, Y) = -\langle (\nabla_X A)Y, a\rangle + \langle A^2 X, Y\rangle f.$$

The Laplacian is the trace of the Hessian. Evaluating at a point p with a geodesic orthonormal frame $\{e_1, e_2\}$ (where $\nabla e_i = 0$ at p), the trace of the covariant derivative commutes with the trace operator, yielding $\mathrm{tr}(\nabla_{e_i} A) = e_i(\mathrm{tr}(A))$. Since M is maximal, $\mathrm{tr}(A) \equiv 0$, making this first term strictly vanish. The trace of the second term gives $\mathrm{tr}(A^2)f$. We finally arrive at:

$$\Delta\langle N, a\rangle = \mathrm{tr}(A^2)\langle N, a\rangle.$$

$\square$

Lemma 3. *With the same definitions, the squared norm of the gradient is:*

$$|\nabla\langle N, a\rangle|^2 = \frac{1}{2}tr(A^2)\left(\langle N, a\rangle^2 + \langle a, a\rangle\right). \tag{12}$$

Proof. From Lemma 1, we have $\nabla f = -A(a^\top)$. Thus:

$$|\nabla f|^2 = g(-A(a^\top), -A(a^\top)) = g(A^2(a^\top), a^\top).$$

Because M is two-dimensional and $\mathrm{tr}(A) = 0$, the endomorphism A possesses eigenvalues λ and $-\lambda$. Hence $A^2 = \lambda^2 I = \frac{1}{2}\mathrm{tr}(A^2)I$. Substituting this into our norm equation gives:

$$|\nabla f|^2 = \frac{1}{2}\mathrm{tr}(A^2)|a^\top|^2.$$

From the decomposition (10), we have $a = a^\top - \langle N, a\rangle N$. Taking the Lorentzian square norm of both sides reveals $|a^\top|^2 = \langle a, a\rangle + \langle N, a\rangle^2$. Substituting this back into the expression for $|\nabla f|^2$ concludes the proof:

$$|\nabla \langle N, a\rangle|^2 = \frac{1}{2}\mathrm{tr}(A^2)\left(\langle N, a\rangle^2 + \langle a, a\rangle\right). \tag{13}$$

□

4 Conformal Metrics and Harmonic Functions

We now apply algebraic choices for the fixed vector $a \in \mathbb{L}^3$ to construct a harmonic function, and subsequently, a flat conformal metric. Both these geometric objects are the key points of Romero's procedure.

4.1 Construction of a Harmonic Function

Choose a *lightlike* vector fixed $a \in \mathbb{L}^3$, so $\langle a, a\rangle = 0$. Because N is necessarily timelike, N and a cannot be orthogonal. Hence, $\langle N, a\rangle$ never vanishes. Up to reversing the sign of a, we assume $\langle N, a\rangle > 0$.

Let us define the function $F = \frac{1}{\langle N, a\rangle}$. We want to evaluate the Laplacian of F:

$$\Delta\left(\frac{1}{f}\right) = \frac{2}{f^3}|\nabla f|^2 - \frac{1}{f^2}\Delta f.$$

Substitute $f = \langle N, a\rangle$. By Lemmas 2 and 3:

$$\begin{aligned}\Delta\left(\frac{1}{f}\right) &= \frac{2}{f^3}\left(\frac{1}{2}\mathrm{tr}(A^2)f^2\right) - \frac{1}{f^2}\left(\mathrm{tr}(A^2)f\right)\\ &= \mathrm{tr}(A^2)\frac{1}{f} - \mathrm{tr}(A^2)\frac{1}{f} = 0.\end{aligned}$$

We have rigorously demonstrated that $\frac{1}{\langle N, a\rangle}$ is a strictly positive **harmonic function** on the maximal surface M.

4.2 Construction of a Conformal Flat Metric

Next, choose a *timelike* vector $b \in \mathbb{L}^3$ such that $\langle b, b\rangle = -1$ and $\langle N, b\rangle > 0$.

Define the function $G = \ln(\langle N, b\rangle + 1)$. To calculate its Laplacian, we employ the rules of calculus alongside the properties derived earlier. Let $v = \langle N, b\rangle$. Then $G = \ln(1 + v)$. The Laplacian can be computed using the chain and quotient rules:

$$\Delta \ln(1+v) = \operatorname{div}\left(\frac{\nabla v}{1+v}\right) = \frac{(1+v)\Delta v - |\nabla v|^2}{(1+v)^2}.$$

Since b is a timelike vector with $\langle b, b\rangle = -1$, Lemma 3 yields $|\nabla v|^2 = \frac{1}{2}\operatorname{tr}(A^2)(v^2 - 1)$. From Lemma 2, we have $\Delta v = \operatorname{tr}(A^2)v$. Substituting these expressions directly:

$$\begin{aligned}
\Delta \ln(1+v) &= \frac{(1+v)\operatorname{tr}(A^2)v - \frac{1}{2}\operatorname{tr}(A^2)(v^2-1)}{(1+v)^2} \\
&= \frac{\operatorname{tr}(A^2)\left[2v(1+v) - (v-1)(v+1)\right]}{2(1+v)^2} \\
&= \frac{\operatorname{tr}(A^2)(1+v)\left[2v - (v-1)\right]}{2(1+v)^2} \\
&= \frac{\operatorname{tr}(A^2)(v+1)}{2(1+v)} = \frac{1}{2}\operatorname{tr}(A^2).
\end{aligned}$$

As shown in our reduced Gauss equation (9), we know that $K = \frac{1}{2}\operatorname{tr}(A^2)$. This directly gives:

$$\Delta G = K. \tag{14}$$

Now we refer to the transformation of the Gaussian curvature under conformal metric changes. According to the general formula derived in class, if a two-dimensional Riemannian manifold (M, g) undergoes a conformal change of metric given by $g^* = e^{2w}g$ for some smooth function $w \in C^\infty(M)$, its new Gaussian curvature K^* satisfies:

$$K - K^* e^{2w} = \Delta w. \tag{15}$$

In our case, the conformal metric is defined as $g^* = (\langle N, b\rangle + 1)^2 g$. We can rewrite this exponentially as $g^* = e^{2\ln(\langle N,b\rangle+1)}g$. By taking our conformal factor to be exactly the function $w = G = \ln(\langle N, b\rangle + 1)$, we can directly apply the transformation formula (15).

Plugging our derived relation (14) ($\Delta G = K$) into the Liouville equation, we obtain:

$$K - K^* e^{2G} = \Delta G = K. \tag{16}$$

Subtracting K from both sides leaves us with $-K^* e^{2G} = 0$. Since the exponential term e^{2G} is always strictly positive, this beautifully algebraic cancellation ensures that:

$$K^* = 0. \tag{17}$$

Therefore, the metric g^* **is strictly flat** everywhere on the maximal graph.

5 Entire Spacelike Graphs and Completeness

Let $u : \mathbb{R}^2 \to \mathbb{R}$ be a smooth, entire function. Its graph defines an immersion mapping $(x_1, x_2) \mapsto (x_1, x_2, u(x_1, x_2))$. In addition, the tangent vectors are $X_1 = (1, 0, u_{x_1})$ and $X_2 = (0, 1, u_{x_2})$. The metric is spacelike precisely when $|Du| < 1$.

5.1 Completeness of the Auxiliary Metric

An essential obstruction in global differential geometry is ensuring that intrinsic lengths of divergent curves go to infinity. Let $\tilde{g} = dx_1^2 + dx_2^2 + dx_3^2$ denote the standard Euclidean metric on the parameter plane $\mathbb{R}^3$, which we know is complete since the plane is complete.

Consider the auxiliary metric given by:

$$g' = \frac{1}{1 - |Du|^2} g, \tag{18}$$

where g is the induced metric of the surface. We claim that this metric is globally complete. Our method relies on systematically comparing the induced metric g to the standard Euclidean metric $\tilde{g} = dx_1^2 + dx_2^2 + dx_3^2$ defined on ambient $\mathbb{R}^3$, identically following the method detailed in class (pages 82–83).

For any tangent vector on the graphical surface $w = aX_1 + bX_2$, where $X_1 = (1, 0, u_{x_1})$ and $X_2 = (0, 1, u_{x_2})$ form a coordinate basis, we can bound the squared length of w with respect to the induced Riemannian metric g:

$$\begin{aligned} g(w,w) &= (a,b) \begin{pmatrix} 1 - u_{x_1}^2 & -u_{x_1}u_{x_2} \\ -u_{x_1}u_{x_2} & 1 - u_{x_2}^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \\ &= a^2 + b^2 - (au_{x_1} + bu_{x_2})^2. \end{aligned}$$

Applying the Cauchy-Schwarz inequality to the second term,

$$(au_{x_1} + bu_{x_2})^2 \le (a^2 + b^2)(u_{x_1}^2 + u_{x_2}^2) = (a^2 + b^2)|Du|^2.$$

Inserting this inequality gives a strong lower bound:

$$g(w,w) \ge (a^2 + b^2) - (a^2 + b^2)|Du|^2 = (1 - |Du|^2)(a^2 + b^2).$$

Simultaneously, the Euclidean norm of this tangent vector in the flat background is evaluated via $\tilde{g}$. Since $w = (a, b, au_{x_1} + bu_{x_2})$, we have:

$$\tilde{g}(w,w) = a^2 + b^2 + (au_{x_1} + bu_{x_2})^2 \le (a^2 + b^2) + (a^2 + b^2)|Du|^2 = (1 + |Du|^2)(a^2 + b^2).$$

Because $|Du| < 1$, we can explicitly state $\tilde{g}(w,w) \le 2(a^2 + b^2)$. Rewriting this, we have $a^2 + b^2 \ge \frac{1}{2}\tilde{g}(w,w)$. We map this directly into our lower bound for $g(w,w)$ to obtain:

$$g(w,w) \ge (1 - |Du|^2)\frac{1}{2}\tilde{g}(w,w).$$

Using our definition for the conformal metric g', this translates to:

$$g'(w,w) = \frac{1}{1 - |Du|^2} g(w,w) \ge \frac{1}{2}\tilde{g}(w,w). \tag{19}$$

Let $\gamma : [0, \infty) \to \mathbb{R}^2$ be a smooth divergent curve in the parameter plane (meaning it escapes every compact subset). Its length $L_{\tilde{g}}(\gamma)$ with respect to the complete metric $\tilde{g}$ satisfies $L_{\tilde{g}}(\gamma) = \infty$.

By integrating the strict inequality $g'(\gamma', \gamma') \ge \frac{1}{2}\tilde{g}(\gamma', \gamma')$ given by (19) over the curve, we obtain the length comparison:

$$L_{g'}(\gamma) = \int_0^\infty \sqrt{g'(\gamma'(t), \gamma'(t))}dt \ge \frac{1}{\sqrt{2}} \int_0^\infty \sqrt{\tilde{g}(\gamma'(t), \gamma'(t))}dt = \frac{1}{\sqrt{2}} L_{\tilde{g}}(\gamma) = \infty. \tag{20}$$

Since every divergent curve has an infinite length $L_{g'}$, by the Hopf-Rinow theorem criterion, g' is a **complete metric**.

It is worth noting that, by a classical result of Nomizu and Ozeki [5], any non-compact Riemannian manifold admits a conformal factor that makes it geodesically complete. The key point in this proof is not just finding a complete conformal metric, but finding one whose conformal factor simultaneously forces the new Gaussian curvature K^* to identically vanish.

6 Proof of the Calabi-Bernstein Theorem

Theorem 1 (Calabi-Bernstein)**.** *The only entire solutions $u = u(x_1, x_2)$ to the maximal surface equation div $\left(\frac{\nabla u}{\sqrt{1-|\nabla u|^2}}\right) = 0$, with $|\nabla u| < 1$, are affine functions $u(x_1, x_2) = ax_1 + bx_2 + c$ satisfying $a^2 + b^2 < 1$.*

Proof. Consider the entire graph represented by a maximal function $u(x_1, x_2)$. Choose the past-directed vector $b = (0, 0, -1)$, giving $\langle N, b\rangle = \frac{1}{\sqrt{1-|Du|^2}} \geq 1$.

Evaluate the conformal metric we built:

$$g^* = (\langle N, b\rangle + 1)^2 g = \left(\frac{1}{\sqrt{1-|Du|^2}} + 1\right)^2 g. \tag{21}$$

Since

$$\left(\frac{1}{\sqrt{1-|Du|^2}} + 1\right)^2 > \frac{1}{1-|Du|^2},$$

g^* dominates g'. Because g' is complete, g^* **is complete**.

By the global version of the Classification Theorem for 2D Riemannian Manifolds of constant curvature, since (M, g^*) is complete, simply-connected and flat, there is a global isometry $\Phi : \mathbb{R}^2 \to M$.

Recall the function $F = \frac{1}{\langle N,a\rangle}$. We have already proven it is positive and harmonic on (M, g), meaning $\Delta_g F = 0$. However, we now reside in the mathematical space of (M, g^*). We must verify that harmonicity is strictly maintained through the conformal rescaling $g^* = e^{2\sigma} g$, adhering to the precise derivation covered in class (pages 86–88).

For an arbitrary function f, we determine how its Laplacian transforms under the metric $g^* = e^{2\sigma} g$. First, we note that the gradients are related by $\nabla_{g^*} f = e^{-2\sigma} \nabla_g f$, since $g^*(\nabla_{g^*} f, X) = X(f) = g(\nabla_g f, X)$ for any vector field X.

Next, applying the Koszul formula to relate the Levi-Civita connections ∇^* and ∇, we have for any vector fields X, Y:

$$\nabla^*_X Y = \nabla_X Y + X(\sigma)Y + Y(\sigma)X - g(X, Y)\nabla_g \sigma.$$

By taking the trace with respect to a local g-orthonormal frame, the relation between the divergences of a vector field Z in dimension $n = 2$ simplifies elegantly to:

$$\mathrm{div}_{g^*} Z = \mathrm{div}_g Z + 2Z(\sigma).$$

We now apply this directly to compute the scaled Laplacian. Since

$$\Delta_{g^*} f = \mathrm{div}_{g^*}(\nabla_{g^*} f) = \mathrm{div}_{g^*}(e^{-2\sigma} \nabla_g f),$$

we can use the standard Leibniz rule for divergence $\operatorname{div}(uZ) = u\operatorname{div}(Z) + Z(u)$:

$$\begin{aligned}
\Delta_{g^*} f &= e^{-2\sigma}\operatorname{div}_{g^*}(\nabla_g f) + (\nabla_g f)(e^{-2\sigma}) \\
&= e^{-2\sigma}\left[\operatorname{div}_g(\nabla_g f) + 2(\nabla_g f)(\sigma)\right] - 2e^{-2\sigma}(\nabla_g f)(\sigma) \\
&= e^{-2\sigma}\Delta_g f.
\end{aligned}$$

This algebraic cancellation demonstrates definitively that if $\Delta_g f = 0$, then necessarily $\Delta_{g^*} f = 0$. Thus, F remains a robustly harmonic function on (M, g^*). Pulling back through our global isometry Φ, $F \circ \Phi$ realizes a positive harmonic function bounded below (since it is strictly positive) on the entire Euclidean plane $\mathbb{R}^2$.

By **Liouville's theorem**, $F \circ \Phi$ (and hence F) must be constant. Thus $\langle N, a\rangle$ is constant.

From Lemma 2, $0 = \Delta\langle N, a\rangle = \operatorname{tr}(A^2)\langle N, a\rangle$. Since $\langle N, a\rangle > 0$, we get $\operatorname{tr}(A^2) \equiv 0$. Because g is positive-definite, $A \equiv 0$, meaning the graph is **totally geodesic**. This is geometrically equivalent to u being an affine function whose gradient strictly satisfies the spacelike bound $|\nabla u| < 1$. □

7 Comments and Generalizations

While classical proofs of the Euclidean Bernstein problem were of an exclusively analytic nature using complex variables, Chern's 1969 article [3] provided a purely geometric proof. Chern defined a flat conformal metric $d\sigma = (1 + 1/W)ds$, demonstrated its completeness, and applied Liouville's theorem directly to the Laplacian. The brilliance of Romero's 1996 paper [7] lies in discovering how to masterfully adapt this exact geometric philosophy of Chern to the Lorentzian setting.

This geometric method avoids boundary behavior problems when mapping to the hyperbolic disk, achieving generalizations into other domains natively.

7.1 Higher Dimensional Generalizations

Historically, one of the crucial turning points of the Euclidean Bernstein problem was its failure past dimension 8. Bombieri, De Giorgi, and Giusti proved that in $\mathbb{R}^9$, there exist non-linear entire solutions to the minimal surface equation [8]. Surprisingly, the Lorentzian analogue is much more rigid. S. Y. Cheng and S. T. Yau [2] proved that the generalized Calabi-Bernstein theorem holds for all dimensions: the only entire maximal spacelike hypersurfaces defined over $\mathbb{R}^n$ inside $\mathbb{L}^{n+1}$ are the linear hyperplanes. Their proof utilizes hard analytical gradient estimations and volume growth measurements, as the uniformization mapping to the Euclidean plane used by Romero is strictly a two-dimensional phenomenon.

7.2 Generalizations to Other Spacetimes

The problem naturally extends to curved ambient spaces, where researchers ask if entire maximal graphs are necessarily totally geodesic. Latorre and Romero [4] studied the Calabi-Bernstein problem in irrotational standard static spacetimes, proving rigidity under certain curvature conditions. Alías and Mira [6] further systematically explored maximal surfaces in more general ambient spaces.

As discussed during the course, one of the most fruitful settings involves Generalized Robertson-Walker (GRW) spacetimes. A GRW spacetime is a product manifold $\overline{M} = I \times M$ endowed with a warped metric $\overline{g} = -dt^2 + f(t)^2 g_M$, where I is an open real interval, (M, g_M) is a Riemannian manifold (representing the spatial fabric), and $f(t) > 0$ is a smooth warping function determining the universe's expansion. In these spaces, under the assumption of the Timelike Convergence Condition (which physically translates to gravity acting strictly as an attractive force), it has been shown that compact maximal surfaces, or those mapped onto entire spacelike slices, exhibit the Bernstein rigidity, squashing down into the totally geodesic slices $t =$ constant.

7.3 Open Problem: Thresholds in GRW Spacetimes

While the Calabi-Bernstein property has been firmly established in flat $\mathbb{L}^3$ and in GRW spacetimes under strict energetic conditions, a compelling **open problem** for future research is to determine the precise geometric threshold where this rigidity fails.

Building upon the elegant mathematical machinery shown above, it remains an open question to explore whether the rigidity holds under generic, unexplored families of the warping function $f(t)$—for example, in scenarios where the spacetime exhibits non-constant scalar curvature, or for spatial fabrics M whose curvature varies locally. Discovering the exact analytical threshold on $f(t)$ that separates the non-existence of non-trivial maximal graphs from the existence of wild, non-flat maximal surfaces would provide profound insights into the mathematics of cosmic censorship and spacetime singularities.

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